

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 26

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) • X survives from about pH 5.5 to 8.5 (wide range); Y only from about pH 6.0 to 8.0 (narrow range) • Both have the same optimum (pH 7.0)	3
01.2	• X: 60% survive at pH 6.0 whereas only 5% of Y survive • X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions • Survivors breed and their offspring are adapted so the population rebuilds; Y is likely to die out locally	3
01.3	• Reveals the habitat requirements / which abiotic factor is near the edge of tolerance • Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival	2
01.4	<i>Any 2 from:</i> • Food supply • Pollinators • Seed dispersal species • Disease / pathogens • Predators / competitors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	

Question	Marking guidance	Marks
02.1	<i>Any 4 from:</i> • Zooxanthellae are photosynthetic algae living inside the polyp cells • The algae absorb CO₂ and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) • They supply the polyp with oxygen and glucose - over 95% of its food energy • In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water • Polyps also feed on plankton captured by cilia and nematocysts	4
02.2	• Corals deposit a skeleton of calcium carbonate (calcite) • Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink	2
02.3	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income • Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
03.1	• Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)	1
03.2	• Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield	2
03.3	<i>Any 2 from:</i> • Replacement of valuable animals that die with genetically identical individuals • Replacement of herds culled during a disease outbreak • Production of large numbers of individuals with desirable characteristics	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
04.1	• Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups	3
04.2	<i>Any 5 from:</i> • Large enough suitable habitat for a viable population • Reliable food supply • Low predation risk / original threat removed • Suitable breeding sites • Water • Support of the local human population • Official / legal protection of habitat and species	5
04.3	• Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed • Identifies causes of failure so future releases can be improved / extra support given	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
05.1	• The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out • Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost	2
05.2	<i>Any 3 from:</i> • Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth • The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding • Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients • Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted	3

Question	Marking guidance	Marks
05.3	• Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances • Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving); specialised flowers reduce pollen being wasted on other species	2
05.4	• Dispersal is not haphazard: animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive • Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling • Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
06.1	• Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora • 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work)	2
06.2	• $N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ • $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ • $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1)	3
06.3	• Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) • Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread	2
06.4	• A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants • Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening	2
06.5	• Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
07.1	• Organic advantages: often waste products, locally available and cheap; add humus improving structure and water retention; increase soil biota; slow release reduces leaching • Organic disadvantages: nutrient composition cannot be controlled; released slowly so needs long-term planning; bulky with high water content so expensive to transport; cannot be applied to a growing crop; odour / pathogens • Inorganic advantages: composition controlled to match crop needs; nutrients released rapidly; concentrated and easy to apply • Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process); no organic matter so humus and biota decline; some toxic to worms; high solubility so leached causing eutrophication; raw materials (phosphate rock) finite	6

Question	Marking guidance	Marks
07.2	• Crops grown in a circulating nutrient solution (inert medium) in greenhouses; nutrient concentrations monitored and adjusted so nutrients are never limiting; abiotic factors controlled • Roots are small (all in contact with nutrients) so more growth goes into the harvested crop; no soil-borne pathogens or weeds; crop stays fresh with roots • Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping); high level of technical knowledge; only economic for high-value crops	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	
08.1	• B	1
08.2	<i>Any 4 from:</i> • Harvesting no faster than the MSY - resource never depleted • Mixed species plantations - more food choices, relationships and niches • Indigenous species - support indigenous insects, birds • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield • Selective logging - small clearings, less disturbance, long-term timber benefit • Retention of dead wood, buffer zones, corridors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
09.1	• Environmental DNA (eDNA) analysis	1
09.2	• Transmitters are too heavy for small birds / their batteries would not last for a long-term study • Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded	2
09.3	• Each separate (regional) population has its own unique gene pool • DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
10.1	• By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t • $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%)	2
10.2	• The nets have a very small mesh size so few non-target animals can escape	1
10.3	• They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young • They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred • Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value	3

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10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing • Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
11.1	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradications, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25